

Final-Year Practical Project Cover Sheet

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Project Title: Modelling the influence of temperature and rainfall on the population dynamics of *Aedes aegypti* over Collier County, Florida, USA

I hereby declare that this Project has not been submitted, either in the same or different forms, to this or any other university for a degree. I also declare that this Project does not draw from any other work prepared under consultancy or other professional undertaking, by myself or jointly with other authors in any way other than that duly and explicitly acknowledged herewith.

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Final-Year Practical Project reports: assessment criteria

These criteria are to be used for all types of final-year practical project reports: *Lab project*, *Field project* and *Computational project*. Outside reading is fundamental when writing up a research project, so it is not mentioned explicitly in the criteria that follow. Most outside reading should be from the peer-reviewed scientific literature, including primary research papers. The expected format of the report is described in detail in the guidelines for **Practical project reports**.

Class	%	Criteria
1 st	100	The report is of sufficient quality to submit for publication to an international peer-reviewed journal (assuming, ideally, that positive and negative results have equal merit).
	95	The report is close to a publishable standard, containing a succinct survey of the most important primary literature and an accurate and logical account and justification of the methods used . It presents the results in a publishable format and knowledgeably applies any necessary mathematical and/or statistical techniques . Discussion of results demonstrates high levels of rigour and critical ability in the context of the relevant literature. Report demonstrates an appreciation of the limitations of the experimental or other procedures, shows attention to detail (in references, figures, etc.), and shows clear and possibly novel insight into the subject.
	90	
	85	
	80	Excellent report, meeting all of the criteria for a mark of 68 and most but not all of the criteria for a mark of 90+.
	76	Excellent report, meeting all the criteria for a mark of 68 and one or a few of the criteria for a mark of 90+.
	72	
2A	68	Very good, well-structured report written in good scientific style (<i>i.e.</i> in clear, concise, direct, precise, dispassionate scientific English, including all important details). It shows the following features: (i) an ability to carry out experimental procedures successfully to generate original results (which may be negative and need not be novel); (ii) a very good understanding of the study design and the methods used to generate and analyse the data; (iii) appropriate – if not high-level – analyses ; (iv) clear presentation of results ; (v) sound knowledge of how the study fits in to the relevant literature ; (vi) some critical interpretation of the results and the study overall.
	65	
	62	
2B	58	Good report showing the following features: (i) an ability to follow experimental procedures ; (ii) basic understanding of the relevant concepts and methods ; (iii) mostly logical structure and scientific style ; (iv) reasonable interpretation of the data or information collected; and (iv) a reasonable attempt to relate the results to the latest literature . Reports that are too long, poorly written , and/or that show poor use of references are unlikely to be marked above a 2B.
	55	
	52	
3 rd	48	Acceptable report showing the following features: (i) an ability to follow some experimental procedures ; (ii) a weak grasp of most of the relevant concepts and methods ; (iii) need for close guidance in design and interpretation ; and (iv) at best limited relation of the results to the relevant literature . Research projects in this bracket are likely to be marred by significant errors, important omissions, brevity and/or a failure to interpret the data critically.
	45	
	42	
Fail	38	Poor report showing the following features: (i) understanding of less than half of the theoretical basis of the project ; (ii) evidence of widespread difficulty following procedures to generate and analyse data; (iii) need for complete instruction in design and interpretation ; (iv) does not relate the outcome of the experimental work to the literature .
	35	
	30	
	25	The report contains more than a few relevant sentences but shows very little understanding of the background to the project, the project design, or the methods used to generate or analyse the data. Students in this bracket are unlikely to have been able to carry out even basic procedures, despite proper instruction.
	20	
	15	The report contains only a few sentences relevant to the subject and does not contain any interpretable results.
	10	
	5	
	0	The report contains nothing relevant or was not submitted.

Practical and Literature Projects: Supervisor assessment criteria for lab/data/field-work or research proposal development

Supervisors are required to provide a **single mark that reflects all aspects of a student's performance** when assessed against the marking criteria. This mark may take into account aspects of the final report as it affects the assessment of overall performance, but independent assessment and provision of a mark for the final report are not required.

Class	%	Criteria
1 st	100	Student worked safely, confidently, diligently, and designed appropriate investigations (lab and/or literature) . Where applicable, the student developed a high level of technical expertise . Student kept supervisor informed of progress, but consistently showed originality and initiative , and did not require any close management. Student contributed strongly to the research in the lab/field or in developing the research proposal . Throughout the project, the student demonstrated a high level of rigour and critical, synthetic and/or analytical ability in the context of the relevant literature and an appreciation of the strengths and limitations of the study design or research project development . Overall, the student performed like a research postgraduate .
	95	
	90	
	85	Student met all of the criteria for a mark of 68 as well as most of the criteria for a mark of 90+
	80	
	76	Student met all of the criteria for a mark of 68 as well as one or a few of the criteria for a mark of 90+ .
	72	
2A	68	Student's lab or field work and research proposal development were performed competently . The student contributed significantly to the experimental and/or research proposal design , worked with some diligence, largely independently , and, where applicable, learned procedures well . The student shows a very good understanding of the study design and the methods used to generate and analyse the data or those proposed as part of a research proposal; they demonstrated sound knowledge of how the study/proposed study fits in to the relevant literature ; and shows some rigour and critical, synthetic or analytical ability in the context of the relevant literature and an appreciation of some of the strengths and/or limitations of the study design or research project development .
	65	
	62	
2B	58	Student's lab, field and/or research proposal development work was performed safely throughout . The student had some input into experimental design and/or research proposal design and worked with some diligence . The student was able to work usefully, requiring only day-to-day supervision . The student shows basic understanding of the relevant concepts and methods and a reasonable ability to place the material in the context of the relevant literature, but only a partial appreciation of the strengths or limitations of the study design or research project development .
	55	
	52	
3 rd	48	The student showed some ability to follow experimental procedures, or to do some research proposal development work, without close supervision and appreciated safety aspects , but the work was limited in quantity or quality . The student's input into experimental design and/or research proposal design was minimal . Student performance in this class is likely to be marred by errors of understanding and a failure to appreciate the strengths or limitations of the study design or research project development .
	45	
	42	
Fail	38	Student worked for up to half of the expected time and worked safely/adequately in the field, lab and/or on the research proposal only when closely supervised . The student showed very little or no initiative or independence . Student performance in this class is likely to be marred by significant errors of understanding and a failure to appreciate the limitations of the study design or research project development .
	35	
	30	
	25	Student attended the laboratory or field site, or meetings with supervisor(s), for up to a third of the expected time and performed some work safely/adequately, but only when closely managed . Very little useful work completed in the field, lab and/or research proposal .
	20	
	15	Student attended the laboratory or field site, or meetings with supervisor(s) , but either attended for less than a quarter of the expected time or worked in an unsafe or otherwise wholly unsatisfactory fashion despite instruction. Negligible amount of work completed in the field, lab and/or on the research proposal.
	10	
	5	Student did not attend the laboratory or field site, or meetings with supervisor(s), or was barred for preventable reasons (e.g., an unacceptable attitude to safety or the work in general) or failed to generate a research proposal.
	0	

Modelling the influence of temperature and rainfall on the population dynamics of *Aedes aegypti* over Collier County, Florida, USA

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ABSTRACT

Mosquito-borne diseases like dengue, chikungunya and yellow fever pose a major global health threat. The *Aedes aegypti* mosquito is a primary carrier of these viruses that thrive in urban environments. Climate change is expanding its range into subtropical regions like Florida, where hot, wet summers and mild, drier winters favour outbreaks. Predicting mosquito populations is vital for disease control, but most mathematical models linking temperature and rainfall to populations were designed for stable tropical climates. Their accuracy in seasonal subtropical zones, such as Florida, was unknown.

This study tested whether a climate-driven ordinary differential equation model developed for *Aedes aegypti* in tropical Brazil could accurately simulate populations in subtropical Collier County, Florida. The model tracks daily changes in the numbers of aquatic and adult female mosquitoes based on local temperature and rainfall. The climate data from Collier County in 2015 was used to run the model and compare its weekly averaged adult female predictions against weekly averaged actual mosquito trap counts from the same year. Model timing and output magnitude were adjusted to optimise the comparison.

The model captured the broad seasonal pattern: low mosquito numbers in cool, dry winter months and higher numbers peaking in the warm, wet summer. However, it failed to accurately predict the exact timing or size of the short-lived peaks observed in the trap data during summer. Statistical comparisons showed only moderate alignment between model predictions and actual counts. Models developed for tropical climates need significant refinement using local data to be effective in subtropical regions with distinct seasons.

1 INTRODUCTION

Globally, vector-borne diseases account for more than 17% of all infectious diseases, causing over 700,000 deaths annually (World Health Organisation, 2024). This pressure is forecasted to intensify in the coming decades as climate change changes the geographical distribution and seasonal activity of disease vectors (Ryan et al., 2019). Among these vectors, *Aedes aegypti* is the primary mosquito species that transmits dengue (Simmons et al., 2012), chikungunya (Leparc-Goffart et al., 2014), and yellow fever (Jentes et al., 2011), which poses a significant public health threat. Unlike other mosquitoes that thrive in rural wetlands, *Ae. aegypti* has high adaptability to urban environments, which evolved to exploit human-dominated environments, breeding in artificial containers such as discarded tyres, flowerpots, and water storage tanks (Kraemer et al., 2015). The close association of *Ae. aegypti* with human habitats, along with its strong preference for feeding almost exclusively on humans during daylight hours and its tendency to rest indoors (Scott & Takken, 2012), has made urban environments into hotspots of disease transmission, particularly in tropical and subtropical regions where warm and humid climates support continuous mosquito activity throughout the year (Ryan et al., 2019).

The expansion of *Ae. aegypti* into new territories is a direct consequence of climate change. Rising global temperatures have extended their potential range polewards, with models predicting that nearly one billion people could face new exposure to this vector by 2080 (Ryan et al., 2019). In the United States, Florida represents one of the most affected regions at the frontline of this vector expansion. The state has a humid subtropical climate, which is characterised by hot summers, mild winters, and abundant rainfall, providing ideal conditions for *Ae. aegypti* proliferation (Stephenson et al., 2022). Over the past two decades, there has been a steady increase in mosquito-borne disease outbreaks in Florida, including autochthonous dengue cases in Key West (2010) and a Zika virus epidemic in Miami-Dade County (2016) that affected over 250 individuals (Centers for Disease Control and Prevention, 2010; Philip, Novick & Novick, 2019). These outbreaks indicate the urgent need to understand how climatic drivers shape *Ae. aegypti* population dynamics through mathematical modelling, which is a prerequisite for effective vector control and outbreak preparedness.

Mathematical modelling approaches variously address these challenges. Empirical statistical approaches describe relationships between environmental variables and mosquito abundance without explicitly modelling underlying biological mechanisms. For instance, a Generalised Additive Model

(GAM) study completed by Brady et al. (2013) has successfully captured complex, non-linear relationships between temperature and adult *Ae. aegypti* survival across diverse laboratory and field settings by flexibly modelling data-driven patterns without predefined mechanistic assumptions, but it lacks explicit biological processes and cannot project population dynamics under novel climate scenarios. By contrast, mechanistic ordinary differential equation (ODE) models offer advantages in that they incorporate biological processes governing mosquito development and survival, enabling explicit simulations of climate effects on mosquito populations. These models are particularly valuable for long-term projections under climate change scenarios, as they can incorporate physiological responses to rising temperatures or shifting precipitation regimes (San Miguel, Da Re & Andreo, 2024).

It is necessary to consider temperature and rainfall while modelling population dynamics, as they are the two most critical climatic factors that regulate *Ae. aegypti*'s life cycle. Laboratory studies demonstrate that temperature regulates key biological processes: egg development ceases below 10°C, larval maturation accelerates between 20–30°C, and adult survival declines sharply above 35°C (Yang et al., 2009). Field observations in Florida further reveal that temperature anomalies can alter mosquito phenology. For example, Hopperstad and Reiskind (2016) analysed 15 years of mosquito surveillance data in Florida and found that mean winter temperatures above 18°C correlated with earlier spring emergence of adult populations. They also found that warmer winter temperatures in southern Florida have been linked to extended seasonal activity of *Ae. aegypti*, potentially increasing arbovirus transmission risk. Meanwhile, rainfall exerts dual effects on *Ae. aegypti* population dynamics. Moderate precipitation creates breeding habitats in containers, while extreme rainfall events can flush larvae from habitats or desiccate eggs during droughts (Morin, Comrie & Ernst, 2013; Nagao et al., 2003). The human water storage behaviours in urban centres in Florida, such as collecting rainwater during dry seasons, make these dynamics more complicated by providing stable breeding sites independent of natural rainfall patterns (Barrera et al., 2008). These climate dependencies suggest that population dynamics are not only region-specific but also sensitive to subtropical climatic variability.

Despite Florida's vulnerability to *Ae. aegypti*-borne diseases, most mechanistic population models have been developed for tropical regions (e.g., Brazil: Cordeiro et al. (2021), Yamashita, Das & Chapiro (2018); Thailand: Dye (1984)). These models rely on parameters derived from lab colonies or field data in stable warm climates (Nagao et al., 2003; Yang et al., 2009, 2016), which perform well in equatorial climates. However, their applicability to subtropical regions such as Florida remains uncertain due to

significant environmental differences, where Florida shows seasonal temperature fluctuations between 15°C to 33°C and irregular rainfall patterns (Florida Climate Centre, 2025), which contrasts sharply with tropical stability. Such disparities may drive distinct biological responses through local adaptation (Bennett, McMillan & Loaiza, 2021), limiting the transferability of tropical models and potentially leading to inaccurate predictions and ineffective disease control in Florida.

To explore this transferability limitation, an ODE model that was developed for *Ae. aegypti* in Lavras, Brazil (Silva et al., 2019), cooperated with climate-dependent functions, and are adapted to Collier County, Florida. Specifically, this study aims to evaluate: to what extent can a Brazil-derived model capture population dynamics in Collier County, Florida? I hypothesised the model would have limited ability to predict the *Ae. aegypti* population in Florida accurately.

2 METHODS

2.1 Mathematical model of mosquito population dynamics

To simulate the population dynamics of *Ae. aegypti* in Florida under the influence of temperature and rainfall, the ODE model proposed by (Silva et al., 2019) and later extended by (Vasconcelos et al., 2022) was adopted. The model describes two compartments: the aquatic stage $A(t)$, which includes eggs, larvae, and pupae, and the adult winged female mosquito population $F(t)$, as they are the only ones that transmit diseases to humans through biting. The governing ODE model equations are Equation 1, where temperature (T) and rainfall (R) represent the two daily independent variables. It is also represented in Figure 1, a compartment model that shows the flow of population.

$$\begin{cases} \frac{dA}{dt} = \varepsilon\varphi(T) \left(1 - \frac{A}{C(R)}\right) F - (\alpha(T) + \mu_1(T, R) + u_1)A, \\ \frac{dF}{dt} = \gamma\alpha(T)A - (\mu_2(T) + u_2)F \end{cases} \quad (1)$$

Table 1 shows the parameter descriptions and their initial values set for the model. The oviposition rate φ is the rate of female mosquitoes' egg laying in the potential breeding sites, and the egg hatching rate ε represents the ability of viable eggs to develop into larvae. Carrying capacity C is interpreted as the overall environmental potential to support the aquatic stages, which is related to the availability of resources such as the abundance of nutrients, space and water that affect aquatic stages. The maturation rate α is the fraction of pupae that become adults. Among all the adult mosquitoes, the proportion of mature individuals that are female is γ . The mortality rate μ of aquatic and adult stages is

represented as μ_1 and μ_2 , respectively. In addition, u_1 and u_2 represent external control measures - the extra mortality caused by larvicides and adulticides, which were set to zero in this study.

Table 1 Parameters and their values for the model of the life cycle.

Parameter	Description	Value	Reference
φ	Oviposition rate	0.5	(Yang, 2017)
ε	Egg hatching rate	0.5	(Yang et al., 2016)
C	Carrying capacity	100-2000	(Yang & Ferreira, 2008)
α	Maturation rate	0.02615 - 0.11612 (days ⁻¹)	(Yang, 2014)
γ	Proportion of female mosquitoes in the population	1.060327 - 8.293997 (days ⁻¹)	(Esteva & Mo Yang, 2005)
μ_1	Aquatic mortality rate	0.01397 - 0.06001 (days ⁻¹)	(Yang, 2014)
μ_2	Adult mortality rate	0.028773 - 0.035859 (days ⁻¹)	(Yang, 2017)
u_1	Mortality rate due to larvicides	0	
u_2	Mortality rate due to adulticides	0	

The non-linear term $(1 - \frac{A}{C})F$ limits the growth of the immature population, which means that as the aquatic population A approaches the carrying capacity C , the number of new eggs laid that successfully become larvae is reduced due to the limited capacity of breeding sites, making the model more biologically realistic.

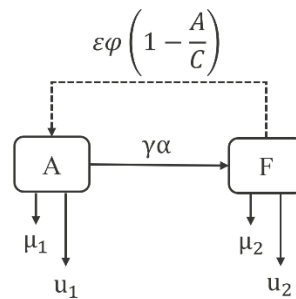


Figure 1 A compartment model of the *Aedes aegypti* population, which illustrates the population flow between the Aquatic stage population (A) and the Female adult population (F).

2.2 Parameterisation according to rainfall and temperature

Temperature and rainfall significantly affect the parameters in the model (Pliego-Pliego et al., 2017; Cordeiro et al., 2021). Therefore, to enhance the model's realism and accuracy, the influence of temperature and rainfall on certain parameters over a unit of time is taken into account, which will be calculated using algebraic expressions (Eq. 2 & 3).

Table 2 Temperature-dependent coefficients calculated by Vasconcelos et al. (2022) using the standard linear least squares methods.

Parameter	b_{0j}	b_{1j}	b_{2j}	b_{3j}	b_{4j}
φ	-5.400	1.800	-2.124×10^{-1}	1.015×10^{-2}	-1.515×10^{-4}
μ_1	2.130	-3.797×10^{-1}	2.457×10^{-2}	-6.778×10^{-4}	6.792×10^{-6}
μ_2	8.692×10^{-1}	-5.723×10^{-2}	1.164×10^{-2}	-1.341×10^{-3}	8.723×10^{-5}
α	1.310×10^{-1}	-5.723×10^{-2}	1.164×10^{-2}	-1.341×10^{-3}	8.723×10^{-5}
	b_{5j}	b_{6j}	b_{7j}		
α	-3.017×10^{-6}	5.153×10^{-8}	-3.420×10^{-10}		

In this ODE model, the parameters that vary with temperature are oviposition rate φ , maturation rate α , aquatic mortality rate μ_1^{temp} , and adult mortality rate μ_2 . They were calculated by using a polynomial function $\theta_j(T)$ at daily average temperature T (°C) (Eq. 2), where $\theta_1(T)$ to $\theta_4(T)$ are φ , α , μ_1^{temp} , and μ_2 , respectively. The parameter is modelled as a polynomial of degree k_j , where b_{ij} are the coefficients of the polynomial terms. The index i ranges from 0 to k_j , which denotes the degree of the polynomial for the function. The coefficients b_{ij} , as shown in Table 2, were obtained from Vasconcelos et al. (2022), where they estimated the values using the standard linear least squares method, based on the experimental data from Yang et al. (2009). This fitting approach aims to minimise the error between observed biological responses and the polynomial curve.

This formulation allows for a flexible, non-linear representation of biological responses to temperature. It captures the complex, often unimodal relationships between temperature and mosquito development rates, such as an optimal temperature range beyond which physiological performance declines.

$$\theta_j(T) = \sum_{i=0}^{k_j} b_{ij} T^i \quad (2)$$

To incorporate the influence of rainfall on mosquito population dynamics, a rainfall-dependent function from Vasconcelos et al. (2022) is adapted. Equation 3 is used to calculate rainfall-dependent model parameters such as the carrying capacity C and the aquatic mortality rate μ_1^{rain} , where they are represented as π_j .

$$\pi_j(R) = \pi_{1j} + (\pi_{2j} - \pi_{1j}) \left(\frac{R - R_1}{R_2 - R_1} \right)^{y_j} \quad (3)$$

In this equation, R represents the daily rainfall, while R_1 and R_2 represent the lower and upper thresholds of rainfall effectiveness throughout the time of study, respectively. In this project, the time of

study is one year long. Rainfall values below R_1 are considered insufficient to sustain aquatic breeding sites, whereas intense rainfall beyond R_2 may result in no further benefit or even habitat disruption, for instance by washing away larvae. The lower and upper bound limits of rainfall will increase larval mortality and hence reduce population growth. The terms π_{1jj} and π_{2j} define the lower and upper bounds of the parameter, at $R = R_1$ and $R = R_2$, respectively. The exponent y_j determines how sensitively the parameter responds to changes in rainfall.

Since the aquatic mortality μ_1 is influenced by both temperature and rainfall (Yang et al., 2016; Sánchez-González et al., 2018; Cordeiro et al., 2021), the additive structure proposed by Vasconcelos et al. (2022) is used by combining the outputs of the temperature-based and rainfall-based parameterisations to represent $\mu_1(T, R)$ (Eq. 4).

$$\mu_1(T, R) = \mu_1^{temp}(T) + \mu_1^{rain} \quad (4)$$

2.3 Numerical Simulation

The simulation of *Ae. aegypti* population dynamics in Florida was conducted using the ODE model (Eq. 1), incorporating climate-dependent functions (Eq. 2-4). R version 4.5.0 was used for all the simulations and statistical analysis, where the ‘deSolve’ package was used to solve the ODE system (Soetaert, Petzoldt & Setzer, 2010). The simulation period covered 365 days, with a time step of 1 day. Both the aquatic-stage population (A) and the female adult population (F) were simulated in the model.

Initial conditions were defined as $A(0) = 200$, $F(0) = 100$, and no control interventions were applied ($u_1 = u_2 = 0$). The initial values for the parameters used in the simulation are shown in Table 1, which were collected from laboratory experiments in Brazil. The temperature-dependent coefficients used are shown in Table 2, which were calculated by Vasconcelos et al. (2022) using standard linear least squares methods. Moreover, the values of optimal parameters y_j in equation 3 are $y_1 = 0.5$ for carrying capacity calculation and $y_2 = 1.98$ for μ_1^{rain} calculation, which were calculated using a Real-Biased Genetic Algorithm (Vasconcelos et al., 2022). The daily average temperature and precipitation climate data in Collier County, Florida, for 2015, were obtained from the Florida Climate Centre, collected from the Naples station.

The carrying capacity C controls density-dependent aquatic growth. This study used a wider range (100–2000) to represent a larger spatial scale, in contrast to the original model’s 10–200.

2.4 Model Validation

To assess the suitability of the model in representing *Ae. aegypti* population trends in Florida, simulation results will be validated against real occurrence data obtained for Collier County. Collier County is in southwestern Florida, which spans 2,305 square miles, with 13.3% of its land covered by freshwater bodies and wetlands (US Census Bureau, 2024). Collier County's climate features a distinct wet season lasting from June to September. Heavy rainfall characterises this period, delivering approximately 90% of the area's annual precipitation. The dry season brings sporadic rain and moderate temperatures. This pronounced seasonal pattern challenges the applicability of tropical mosquito models. Collier County maintains a robust mosquito surveillance system, with adult mosquito surveillance data available from 2010 to 2020. Data obtained from the year 2015 will be used for comparison (Samuel et al., 2023). The occurrence data was collected by using the New Jersey Light Traps (NJLTs) baited with octenol, due to the limited data collected from BG-Sentinel Traps and CDC Light Traps.

The number of mosquitoes captured at different locations on the same day was added together and then divided by two, due to the mixed sexes being collected. In this project, a sex ratio of 50% female adults is assumed as the mean sex ratio of 19 *Ae. aegypti* strains are 19-52% females, as shown by Craig, Vandehey & Hickey (1961). Temperature has a negligible effect on *Ae. aegypti*'s sex ratio, as it is mainly determined by genotype (Hickey, 1970).

Occurrence data and model outputs for female adult mosquitoes F were grouped into epidemiological weeks using the function `isoweek`, due to the absence of occurrence data for certain dates. The data from the same week were averaged to avoid certain extreme values, resulting in average daily occurrence data in different weeks.

Using climate data from Collier County and mosquito surveillance data from the year 2015, model outputs are compared against observed adult abundances. Statistical metrics, such as Root Mean Square Error (RMSE), Pearson's correlation (r), and Spearman's rank correlation (ρ), are used to quantify model performance.

2.5 Model curves adjustment

To improve the agreement between the simulated and observed *Ae. aegypti* female populations, a post-hoc model curve adjustment was applied based on the methodology proposed by Vasconcelos et al. (2021) (Eq. 5). The observed weekly abundance data is denoted by $I(k)$, which were derived from the trap collections and averaged across days for each epidemiological week k . N is the number of epidemiological weeks, which is 53 in the year 2015. The adjustment involved shifting the simulated female adult population $F(k)$ from the ODE model by a time lag τ identified using cross-correlation analysis and scaling the shifted simulated series by an optimal factor λ that minimised the RMSE value. Considering the scaling factor is necessary due to the reason that the observed data is the captured abundance from the traps, instead of the whole population number in Collier County. As a result, the adjusted simulated data was defined using Equation 6.

$$\lambda = \frac{\sum_{k=1}^N I(k) \cdot F(k)}{\sum_{k=1}^N F(k)^2} \quad (5)$$

$$F_{adj}(k) = \lambda \cdot F(k - \tau) \quad (6)$$

The adjusted RMSE, Pearson, and Spearman correlation coefficients were calculated again using the final fit simulated data. This adjustment improves model performance in terms of timing and magnitude when compared to the original simulation output.

3 RESULTS AND DISCUSSION

3.1 Model validation results

Corresponding to Florida's subtropical temperature and precipitation patterns (Fig. 2), the climate-driven ODE model successfully produced known seasonal patterns with low mosquito abundance in the cooler, drier months, and highest in August (Fig. 3). These patterns are consistent with warmer temperatures and summer rainfall jointly promoting faster larval development, increased oviposition rates, and greater availability of breeding sites, driving the distinct seasonal peaks. The aquatic-stage abundance shows a higher variability compared to the adult population, which is consistent with the model's structure, where aquatic mortality is more sensitive to rainfall fluctuations. Aquatic-stage abundance also peaks slightly earlier than the adult female peak. The lagging response to the adult population is biologically plausible, reflecting the time needed for development from larvae to adults. After October, both compartments decline due to cooling temperatures and reduced rainfall.

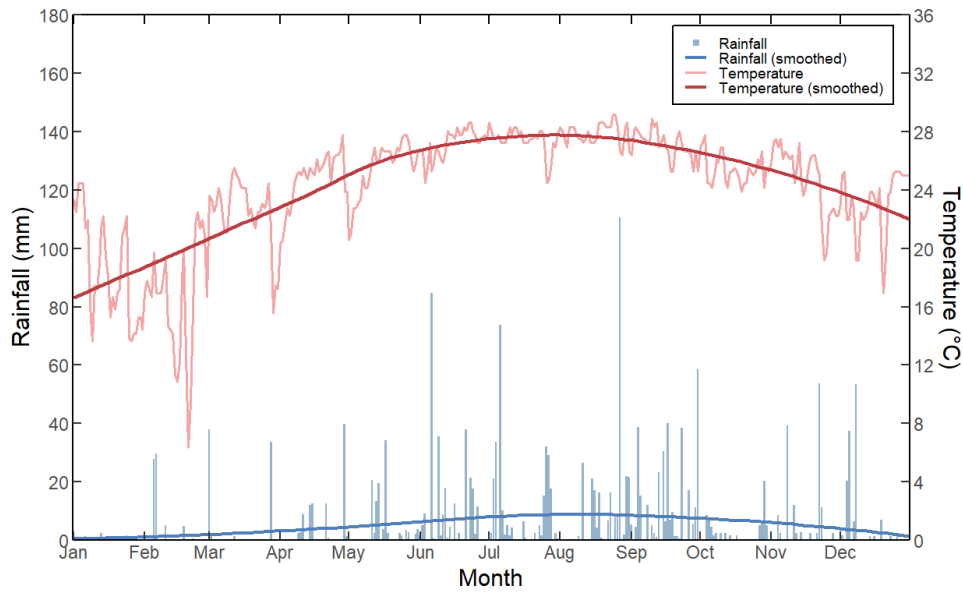


Figure 2 Daily temperature (°C) and rainfall (mm) in Collier County, Florida, for the year 2015. Raw data are shown in light red (temperature) and light blue (rainfall), while LOESS-smoothed trends are shown in dark red and dark blue, respectively. The right-hand y-axis represents temperature, while the left-hand y-axis represents rainfall.

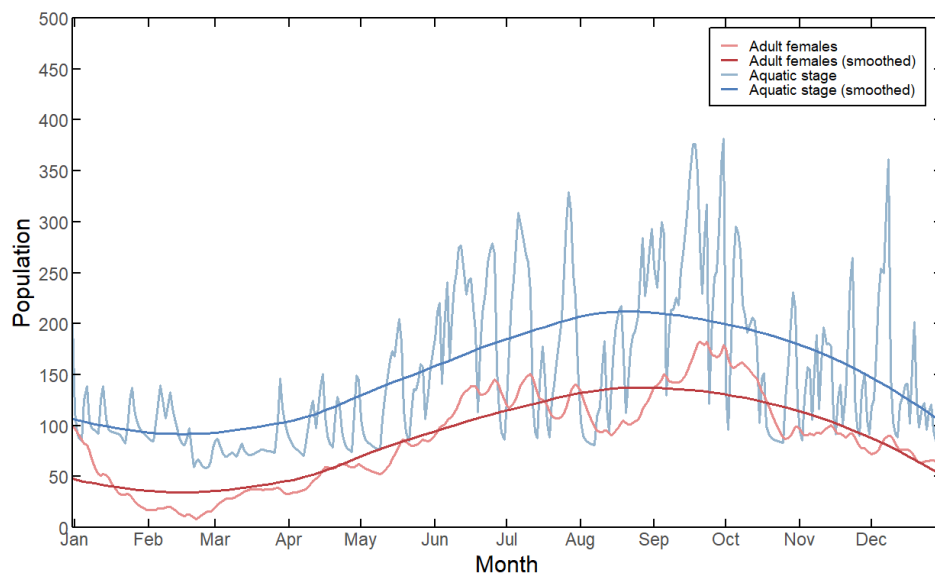


Figure 3 Simulated daily population dynamics of *Aedes aegypti* in Collier County in 2015. Aquatic-stage individuals are shown in blue (raw and smoothed), while adult females are shown in red.

However, the model only captured a broad seasonal trend of *Ae. aegypti* population dynamics in Collier County, with low winter abundance and distinct summer peaks (Fig. 4), even after applying model curve adjustments (Table 3). Its limited ability to predict short-term fluctuations is evidenced by the relatively high root mean square error (1.65) and low correlation coefficients ($r = 0.29$; $\rho = 0.30$),

revealing that mechanistic models parameterised with tropical climate response functions face inherent limitations when applied to Florida’s divergent subtropical conditions. These findings support the hypothesis that the model has limited ability to predict the *Ae. aegypti* population in Florida.

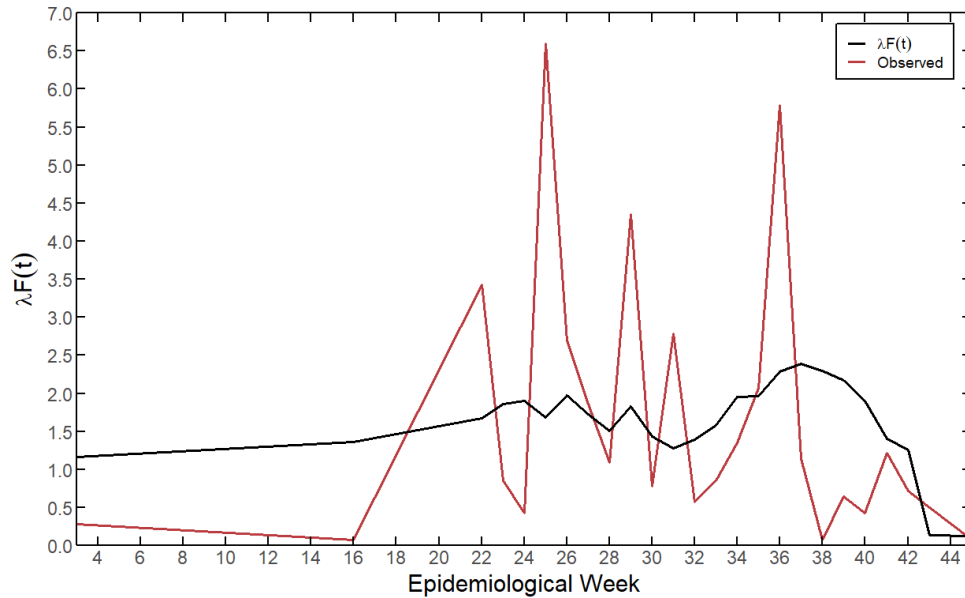


Figure 4 Model validation using observed female *Aedes aegypti* abundance in Collier County. Observed weekly means from trap data (red line) are compared to adjusted model output (black line), where λ is a scaling factor and $\tau = -2$ is the optimal lag based on cross-correlation.

Table 3 Summary of model fit statistics. These statistics demonstrate the fitness between the weekly averaged observed data and simulated data after the adjustments by considering time lag and scaling factor.

Metric	Values
RMSE	1.65
Pearson correlation (r)	0.2922
Spearman correlation (ρ)	0.3042
Optimal lag (τ)	-2 weeks
Scaling factor (λ)	0.01369

The underperformance of the model in predicting weekly abundance peaks during summer suggests that biological parameters calibrated in tropical Brazil inadequately represent mosquito responses to Florida’s climate. Specifically, the minimal temperature in winter and irregular dry season rainfall patterns likely drive local adaptations in development rates and breeding behaviour not captured in models optimised for stable tropical conditions (Bennett, McMillan & Loaiza, 2021). This parameter mismatch represents a core theoretical limitation, although there are other methodological constraints and data limitations to produce the observed prediction errors shown in the following sections.

3.2 Model Design and Assumption Constraints

While the model captured broad patterns, it demonstrated limited accuracy in predicting complex real-world scenarios. Specifically, it is unable to fully capture week-to-week variability, underestimating sharp peaks around weeks 24, 27, and 30. This difference arises partly because the model was designed for generalised environments and ignores specific field conditions like localised breeding sites or trap spatial distribution for data collection. Future models could incorporate spatial heterogeneity using spatially explicit compartments or geospatial data, such as land use maps and breeding site distributions (Lambrechts et al., 2011; Manore et al., 2017). This enables finer-scale simulations and improved correspondence with field observations by resolving localised habitat effects.

Additionally, the model cannot respond to sudden changes driven by microclimatic variations, human activities, or mosquito control efforts. For instance, observed data spikes likely reflect the intense local breeding triggered by short-term rainfall and optimal high temperatures, which is a phenomenon documented in urban mosquito populations (Lambrechts et al., 2011). To improve the simulation of abrupt abundance changes, the model could integrate high-resolution weather data or stochastic elements, such as optimal control elements, including timing and intensity of larvicide or adulticide deployment (Vasconcelos et al., 2022). This would enhance the model's utility for public health planning, particularly in resource-limited settings requiring cost-effective strategies.

Human-mediated habitat creation, such as water storage during droughts may over-weigh the natural precipitation patterns (Flaibani et al., 2020), which may contribute to the causes of the divergence between simulated and observed aquatic stage abundance during rainfall fluctuations (Fig. 4). Specifically, the simulated abundance from the model shows significant sensitivity to rainfall deficits, whereas field data suggests larval populations are sustained during dry periods by artificial water storage through the containers. This indicates that human behaviour decouples actual larval population dynamics from natural precipitation patterns in ways not captured by the current model framework.

In addition, areas with various degrees of development into urban environments may show different levels of human-mediated habitat modification. In this case, the ODE model was developed for Lavras in Brazil, which is a primarily urban area (Sousa et al., 2022), whereas Collier County in Florida is a mixture of both rural and urban environments, which may create fewer artificial breeding sites (Collier County Growth Management, 2020). Therefore, future iterations should incorporate parameters

representing anthropogenic water sources to better reflect mosquito breeding site availability independent of rainfall (Flaibani et al., 2020).

Another issue is the lack of uncertainty quantification. The deterministic nature of the ODE model and its fixed parameters precludes confidence or prediction intervals, making it unable to assess the prediction coverage of the output (Kreutz, Raue & Timmer, 2012). Incorporating techniques such as Monte Carlo simulations (Krausch et al., 2019) or bootstrapping (Joshi, Seidel-Morgenstern & Kremling, 2006) in the dynamical system would be a valuable direction for future work.

Apart from that, the model relies on a fixed scaling factor ($\lambda = 0.0137$) to align simulations with trap counts, assuming constant trap efficiency, ignoring real-world variations due to environmental factors or trap placement. To mitigate this limitation, future models could utilise dynamic or trap-specific scaling factors that account for variation in trap efficiency across time and location, calibrated across different field conditions or using multiple trap types and cross-validating their catch rates.

Although the aquatic mortality rate μ_1 can be calculated accurately using a multivariate function of rainfall and temperature using a Taylor series expansion, this study used a simplified additive approach where μ_1 is the sum of two independent functions to avoid overfitting and focuses on seasonal trends. Future studies should consider using the original multivariate formulation if sufficient empirical data on such interactions becomes available for Florida.

3.3 Data Quality Constraints

Low correlation metrics and the high adjusted RMSE may partly reflect the limitations of the observation data, not solely the inadequacies of the model. The incomplete daily records of the surveillance dataset make it necessary to compute weekly averages for validation. This reduces temporal resolution and weakens apparent model performance. To improve future model evaluation, more consistent and high-frequency mosquito monitoring efforts are needed, ideally using trap types optimised for *Ae. aegypti*, such as BG Sentinels traps (Maciel-de-Freitas, Eiras & Lourenço-de-Oliveira, 2006). Additionally, integrating multiple data sources, such as citizen science records, spatially disaggregated trap networks, or egg count data from ovitraps, could help provide a more complete picture of mosquito abundance and enable finer-scale validation of model outputs.

Moreover, biological parameters derived from laboratory studies introduce external validity concerns, as they ignore field realities like predation, competition, microclimatic variation, and

anthropogenic disturbance that affect survival and behaviour. For example, temperature-dependent maturation and mortality rates were measured under more controlled and stable laboratory conditions compared to the field environments. Meanwhile, a fundamental assumption of this study is that the parameter values estimated for Lavras, Brazil, are transferable to Collier County, Florida. This ignores ecological and social differences, which potentially drive local mosquito adaptations (Bennett, McMillan & Loaiza, 2021). The importance of producing site-specific mosquito entomological data was evident in Brady et al. (2013). Therefore, regionally calibrated parameters from Florida-specific field studies are essential to reduce model misspecification.

The use of surveillance data from Collier County alone may not accurately represent the whole of Florida. Future studies could validate the model by integrating data from other Counties over multiple years, extending simulation periods and covering areas. Incorporating years with extreme weather events like hurricanes or droughts would improve model validation under atypical conditions and enhance understanding of population resilience under future climate scenarios.

3.4 Limitations That Primarily Affect Absolute Abundance Estimation

Three other limitations primarily impact absolute abundance estimation but minimally affect seasonal trend validation after adjusting the data with the scaling factor in this trend-focused study. However, accurate absolute abundance prediction becomes essential for future applications such as informing spatially explicit vector control programmes, refining epidemiological risk assessments, and enhancing the strategic allocation of public health resources (Eisen & Moore, 2013).

Firstly, the model assumes a fixed 1:1 sex ratio in adult emergence. Empirical studies showed a significant variation in sex ratios of *Ae. aegypti*, from 19% to 52% female (Craig, Vandehey & Hickey, 1961). This assumption biases observed female abundance systematically, which indirectly makes the abundance estimation inaccurate while validating the abundance between groups. Future work could incorporate dynamic or empirically derived sex ratios based on local Florida *Ae. aegypti* strains.

Secondly, NJLTs baited with octenol likely underestimates true *Ae. aegypti* abundance due to this species' weak attraction to light-based traps and limited response to octenol compared to human-specific kairomones (Maciel-de-Freitas, Eiras & Lourenço-de-Oliveira, 2006). Field study shows that BG-Sentinel traps outperform NJLTs by 7.9 times in capturing *Ae. aegypti* because they mimic human hosts through visual cues, convection currents, and attractants (Maciel-de-Freitas, Eiras & Lourenço-de-

Oliveira, 2006). Standardised deployment protocols are critical to reduce spatial sampling bias.

Thirdly, there is an uncertainty in defining the true carrying capacity C for *Ae. aegypti* at the scale of Collier County. The chosen range of C in this study, from 100 to 2000, is a broad approximation that lacks empirical validation, influenced by unquantified factors like land use, urbanisation, water storage, and vector control coverage affecting breeding site density and quality. Future studies should estimate area-specific carrying capacities using remote sensing data, field surveys of breeding containers, or models linking land cover and infrastructure to estimate larval habitat availability.

4 CONCLUSION

In conclusion, this study adapted a tropical mechanistic model to simulate climate-driven *Ae. aegypti* population dynamics in subtropical Florida. While capturing the general seasonal abundance trend in Collier County during 2015 after temporal adjustments and scaling, the model demonstrated limitations in predicting short-term peaks and absolute population sizes. While there were several limitations to this approach, the model provides a crucial foundation. More broadly, the methodology offers a blueprint for adapting mechanistic models to other regions where *Ae. aegypti* is spreading due to climate change. Future refinements incorporating spatial heterogeneity, region-specific parameters, anthropogenic water sources, uncertainty quantification, improved surveillance, and multi-year simulations will enhance utility for public health planning.

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APPENDIX

All raw data and code used in this study are available on GitHub: <https://github.com/yukiwxq/Aedes-aegypti-population-dynamics->.